

Prashanth Sai Santosh Koripalli

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Education

Master of Science in Artificial Intelligence

Jan 2024 – Dec 2025

San Jose State University, USA

Course Work: AI and Data Engineering, Machine Learning, Intelligent Autonomous Systems, Deep Learning, Reinforcement Learning, Graphs in Robotics, Advance Data Mining, Enterprise Application Development

Technical Skills

Languages: Python, C++, C, Java, R, Rust, CUDA, Bash, JavaScript, ShellScript, TypeScript, SCSS, HTML/CSS, SQL.

Frameworks: PyTorch, TensorFlow, Ollama, Docker, Kubernetes, React, Next.js, Node, Redux, GraphQL.

Tools/ Paradigms: WandB, Git, Github, Jira, Dspy, OpenAI Gym, Jenkins, Power BI, Tableau, Amazon Web Services.

Databases: MySQL, OracleDB, SQL, PostgreSQL, MongoDB, Firebase, Apache Kafka, Redis.

Experience

Software Developer | Oracle Corporation

Aug 2022 – Jan 2024

- Built scalable ML inference pipelines to transition models from AWS to OCI, and enhanced OCI dashboards to support real-time risk ranking and visualization, improving decision-making for customers like the Red Bull Racing and Pfizer.
- Integrated Newmetrix's AI-powered safety risk models into Oracle Construction Intelligence Cloud (CIC) post-acquisition, enabling predictive analytics and automated safety insights for construction clients.
- Optimized PL/SQL scripts and SQL queries for Primavera Analytics, resolving Data Warehousing issues with sources like P6, Unifier, and tools like OBI and OCI, improving query speed and system reliability by 20%.

Data Analyst Intern | Oracle Corporation

Jan 2022 – July 2022

- Acquired expertise in Oracle tools like Primavera, P6, and OCI within 3 months by actively engaging in training, concurrently working on tasks, and manually testing daily runs, resulting in enhanced tool reliability and project efficiency.
- Automated daily and weekly test runs using Selenium for Oracle's analytics platforms, streamlining UI/UX workflows and functional testing, reducing manual testing time by 50% and improving testing efficiency.
- Led the successful migration of 15,000 Confluence documentations from CEGBU to GBU, ensuring no data loss by developing an automated script to validate and cross-check content, resulting in streamlined access and better collaboration across teams.

Projects

SAMMY AI Chatbot | Python, NLP, RAG, LLM

- Built 'Sammy,' an LLM-powered retrieval system using RAG for SJSU, providing 24/7 responses with 100% positive feedback.
- Optimized inference to support 20 users on a V100 with 200ms latency, scaling to 500 via distributed inference on 8 A100s.
- Engineered a confidence-based response modulation system, dynamically adjusting the chatbot's tone to ensure contextual sensitivity and reducing 99% of model hallucinations in query responses.

PinRank: Intent-Aware Ad Ranking System | PyTorch, Faiss, BERT, CLIP, XGBoost, MTML, YouTube Architecture

- Designed a two-stage recommender system inspired by YouTube's retrieval + ranking architecture and Meta's MTML framework for personalization.
- Implemented a Faiss-based dense retrieval model using multimodal query-ad encoders (BERT for text, CLIP for images) to capture user intent.
- Trained an XGBoost ranker on user interaction logs using pairwise ranking loss, multi-task targets (CTR, NDCG, dwell time).
- Improved NDCG@10 by 20% and CTR by 18% over baseline in offline evaluation using ablation and embedding distillation.

EBay Challenge Entity Extraction (Team Name: Edgerunners) | Python, Ollama, Dspy, PyTorch, TorchTune

- Fine-tuned Llama 3.1 with LoRA to map 5M+ automobile part records to compatible car models in Year-Make-Model format, utilizing 300GB of website data for improved accuracy.
- Designed a distributed inference system across multiple GPU nodes on HPC, cutting inference time from 10 days (single A100 GPU) to 23 hours using 16 A100 GPUs.
- Applied entropy-based sampling to filter high-confidence predictions, increasing the F0.2 score by 13% (0.49 to 0.62).

CTRL-ALT-AMZ: Personalized Product Recommender | PyTorch, FastAPI, FAISS, Amazon Reviews '23

- Built a multimodal recommender using GRU4Rec and transformer-based session modeling for user-item interaction sequences.
- Encoded user behavior with BERT and fused CLIP (images) + BERT (text) embeddings for personalized ad-item re-ranking.
- Achieved 27% boost in Recall@10 over baseline models and improved click-through-rate on product recommendation tasks.

Achievements

- **Achieved 5th Rank in eBay Machine Learning Challenge (2024), USA:** Successfully tackled entity extraction for mapping 5M+ automobile part records to compatible car models.
- **Massachusetts Institute of Technology Covid Challenge, Winning Team (2020):** Won the hackathon by developing a digital loan approval system for unregistered MSMEs to address key challenges faced by small businesses.